

SOLOV'YEVSK, Russian Federation

WMO: 309670

Lat: 49.900N

Lon: 115.750E

Elev: 2044

StdP: 13.64

Time Zone: 9.00 (E09)

Period: 86-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.9	-22.5	-31.0	1.0	-26.5	-26.8	1.3	-22.0	26.9	-1.2	21.6	2.8	2.1	160	0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	85.5	64.3	82.1	62.9	78.9	61.7	68.0	78.7	66.4	75.8	64.9	73.8	8.3	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	99.5	71.3	63.3	93.9	70.3	61.6	88.5	69.4	33.7	78.9	32.4	76.0	31.2	73.9	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	22.4	18.6	-34.0	91.5	5.9	2.7	-38.2	93.5	-41.7	95.1	-45.0	96.6	-49.3	98.5
			-34.1	71.6	5.9	2.6	-38.3	73.5	-41.8	75.0	-45.0	76.5	-49.3	78.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.7	-8.4	0.9	17.8	36.5	51.3	62.2	67.4	64.6	50.8	35.6	13.3	-1.4
	DBStd	27.42	7.70	8.23	10.78	8.03	8.64	7.00	4.69	4.75	7.11	8.24	10.32	8.11
	HDD50	7903	1810	1374	997	409	89	5	0	0	74	449	1102	1594
	HDD65	11989	2275	1794	1462	855	434	129	27	65	427	910	1552	2059
	CDD50	1598	0	0	0	4	128	372	540	451	99	4	0	0
	CDD65	208	0	0	0	0	8	46	101	51	2	0	0	0
	CDH74	2047	0	0	0	0	204	533	801	478	27	4	0	0
	CDH80	506	0	0	0	0	49	129	193	132	3	0	0	0
Wind	WSAvg	6.8	4.6	5.1	7.0	9.6	8.9	8.2	6.7	5.9	7.3	6.9	6.4	5.0
Precipitation	PrecAvg	11.8	0.1	0.1	0.2	0.4	1.0	2.0	3.2	2.6	1.2	0.5	0.2	0.2
	PrecMax	22.7	0.4	0.4	0.9	1.3	2.1	3.9	8.5	6.7	3.0	1.3	0.8	0.6
	PrecMin	7.2	0.0	0.0	0.0	0.1	0.2	0.4	0.6	0.5	0.2	0.1	0.0	0.0
	PrecStd	3.5	0.1	0.1	0.2	0.3	0.5	0.9	1.7	1.5	0.7	0.3	0.2	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.3	33.9	55.1	69.5	86.3	88.0	90.9	88.8	78.2	68.2	43.8	23.3
		MCWB	16.4	28.0	40.3	49.7	60.5	63.2	67.9	67.7	58.9	49.9	34.3	20.5
	2%	DB	11.2	24.3	46.1	61.7	80.0	83.7	85.3	83.9	72.4	58.9	37.4	18.7
		MCWB	9.5	21.1	34.8	44.8	57.6	61.2	66.1	64.9	57.1	44.8	30.2	16.3
	5%	DB	6.9	18.9	39.8	56.8	74.3	79.7	81.8	79.2	67.6	53.8	32.2	14.3
		MCWB	5.5	16.2	30.8	41.7	54.3	60.7	63.9	63.5	55.1	42.2	27.4	12.4
	10%	DB	3.6	15.3	34.8	52.0	68.4	76.5	78.7	75.3	63.8	49.1	28.1	10.7
		MCWB	2.5	13.2	28.3	38.8	50.8	58.9	63.4	61.7	53.0	39.2	23.9	9.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.4	27.9	40.7	49.4	62.6	66.3	70.3	71.0	62.0	50.5	34.7	20.7
		MCDB	18.7	34.1	54.6	66.8	82.7	80.0	82.2	83.4	71.8	67.9	43.2	23.1
	2%	WB	9.5	21.1	35.2	45.0	58.4	64.5	68.2	68.2	59.3	45.5	30.5	16.4
		MCDB	11.1	24.1	45.8	61.0	77.5	77.7	79.7	77.9	67.5	57.8	36.1	18.7
	5%	WB	5.5	16.4	31.2	41.9	54.9	63.0	67.0	66.1	57.0	42.6	27.3	12.5
		MCDB	6.6	18.9	39.1	55.5	72.1	75.7	77.0	74.7	64.8	52.0	31.9	14.2
	10%	WB	2.6	13.2	28.3	39.3	52.1	61.3	65.7	64.3	54.5	40.0	24.1	9.3
		MCDB	3.5	15.3	34.9	50.8	66.7	73.1	74.2	72.4	63.0	48.0	27.9	10.6
Mean Daily Temperature Range	5% DB	MDBR	18.6	23.2	22.9	23.6	24.9	21.1	18.2	18.2	18.6	20.4	18.8	17.7
		MCDBR	21.7	27.6	27.8	32.0	33.4	26.6	24.7	25.6	24.4	28.5	23.1	21.0
		MCWBR	20.1	24.0	18.4	18.0	15.7	10.6	9.2	10.7	12.0	16.7	17.0	18.8
	5% WB	MCDBR	21.5	27.3	26.6	30.1	30.2	22.8	18.8	20.1	19.5	27.2	22.8	20.5
		MCWBR	20.1	23.7	18.1	17.6	15.0	10.1	8.2	10.3	10.4	16.3	16.8	18.5
Clear-Sky Solar Irradiance	taub	0.254	0.280	0.320	0.404	0.484	0.463	0.423	0.389	0.340	0.310	0.278	0.262	
	taud	2.239	2.196	2.189	2.072	1.893	2.078	2.267	2.336	2.411	2.394	2.280	2.186	
	Ebn at Noon	252	272	279	265	247	254	264	268	272	260	239	226	
	Edh at Noon	23	30	37	47	59	49	41	36	30	25	22	21	
All-Sky Solar Radiation	RadAvg	539	918	1390	1721	1953	2041	1884	1685	1346	916	547	404	
	RadStd	29	46	75	107	108	153	121	110	83	42	28	29	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	-4.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page